C Programming Lab Record

Practical Assessment Tasks

STUDENT NAME	REGISTER NUMBER	COURSE / SLOT
P Parvash khan	192312552	CSA 0210

Task 1: Calculate Electricity Bill

Task Description: Write a C program to calculate an electricity bill using household number, consumer category, previous meter reading, and current meter reading. Display units consumed, energy charge, fixed charge, and total bill while validating inputs.

PROGRAM SOURCE CODE

```
#include <stdio.h>

int main() {
    int householdNumber, category;
    long long previousReading, currentReading, units;
    double energyCharge, fixedCharge, totalBill;
    double rate;

    printf("Enter household number: ");
    scanf("%d", &householdNumber);

    printf("Enter category (1-Domestic, 2-Commercial): ");
    scanf("%d", &category);

    printf("Enter previous meter reading: ");
    scanf("%lld", &previousReading);

    printf("Enter current meter reading: ");
    scanf("%lld", &currentReading);

    if (previousReading < 0 || currentReading < previousReading) {
        printf("Invalid meter readings.\n");
        return 1;
    }

    if (category == 1) {
        rate = 5.0;
        fixedCharge = 50.0;
    } else if (category == 2) {
        rate = 8.0;
        fixedCharge = 100.0;
    } else {
        printf("Invalid category.\n");
        return 1;
    }

    units = currentReading - previousReading;
    energyCharge = units * rate;
    totalBill = energyCharge + fixedCharge;

    printf("\n--- Electricity Bill ---\n");
    printf("Household Number: %d\n", householdNumber);
    printf("Units Consumed: %lld\n", units);
```

Task 2: Choose the Data Types & Handle Register Rollover

Task Description: Select suitable C data types for meter readings and calculate electricity units consumed, including handling overflow/rollover when a 16-bit unsigned meter register resets back to zero.

PROGRAM SOURCE CODE

```
#include <stdio.h>
#include <limits.h>

typedef unsigned short meter_t;

unsigned int consumption(meter_t previous, meter_t current)
{
    return (meter_t)(current - previous);
}

int main(void)
{
    printf("sizeof(meter_t) = %zu bytes, maximum reading = %u\n",
           sizeof(meter_t), USHRT_MAX);

    printf("Consumption: %u units\n", consumption(1200, 1530));
    printf("Consumption: %u units\n", consumption(65400, 120));

    return 0;
}
```

EXECUTION OUTPUT

```
sizeof(meter_t) = 2 bytes, maximum reading = 65535
Consumption: 330 units
Consumption: 256 units
```

Task 3: Define the Constants for Tariff Rates

Task Description: Define electricity tariff slab values and fixed charges as constants (`#define`) so that tariff revisions can be updated in a single location in the code.

PROGRAM SOURCE CODE

```
#include <stdio.h>
#include <stddef.h>

#define SLAB1_UNITS 100
#define RATE1_PAISE 0
#define RATE2_PAISE 225
#define RATE3_PAISE 450
#define FIXED_CHARGE_PAISE 5000

long compute_bill(int units)
{
    long paise = FIXED_CHARGE_PAISE;

    if (units <= SLAB1_UNITS) {
        paise += units * RATE1_PAISE;
    } else if (units <= 200) {
        paise += SLAB1_UNITS * RATE1_PAISE;
        paise += (units - SLAB1_UNITS) * RATE2_PAISE;
    } else {
        paise += SLAB1_UNITS * RATE1_PAISE;
        paise += 100 * RATE2_PAISE;
        paise += (units - 200) * RATE3_PAISE;
    }

    return paise;
}

int main(void)
{
    int tests[] = {0, 100, 150, 200, 350};
    int n = sizeof(tests) / sizeof(tests[0]);

    for (int i = 0; i < n; i++) {
        long p = compute_bill(tests[i]);
        printf("%d units -> Rs %ld.%02ld\n", tests[i], p / 100, p % 100);
    }

    return 0;
}
```

EXECUTION OUTPUT

```
0 units -> Rs 50.00
100 units -> Rs 50.00
150 units -> Rs 162.50
200 units -> Rs 275.00
350 units -> Rs 950.00
```

Task 4: Use Macros Safely

Task Description: Write C macros for repeated calculations using safe parenthesization, demonstrate expansion without side effects, and identify safe usage patterns.

PROGRAM SOURCE CODE

```
#include <stdio.h>

#define RUPEES_TO_PAISE(x) ((x) * 100)
#define SQUARE(x) ((x) * (x))
#define MAX(a, b) (((a) > (b)) ? (a) : (b))

int main(void)
{
    int a = 5, b = 8;

    printf("Rupees to paise: %d\n", RUPEES_TO_PAISE(25));
    printf("Square of 3 + 1: %d\n", SQUARE(3 + 1));
    printf("Maximum value: %d\n", MAX(a, b));

    return 0;
}
```

EXECUTION OUTPUT

```
Rupees to paise: 2500
Square of 3 + 1: 16
Maximum value: 8
```